

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12413052
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Xiao; Yong et al.

Electric energy safety authentication apparatus

Abstract

Provided is an electric power safety authentication device, including a front casing, a rear casing, a sealing element, and a fastener connection structure. A sealing groove that matches the sealing element is provided at connection between the front and rear casings. The fastener connection structure comprises a first connector, a second connector, a first fastener, a second fastener, a first spring, and second springs. The first fastener is arranged on the front casing, and the second fastener is arranged on the rear casing. The first fastener and the second fastener each comprise a first clamp and a second clamp. The first connector is fit in and fastened to the first fastener, and the second connector is fit in and fastened to the second fastener.

Inventors:	Xiao; Yong (Guangdong, CN), Zhao; Yun (Guangdong, CN), Lin; Weibin (Guangdong, CN), Cui; Chao (Guangdong, CN), Cai; Ziwen (Guangdong, CN), Lu; Yuxin (Guangdong, CN)
Applicant:	ELECTRIC POWER RESEARCH INSTITUTE, CHINA SOUTHERN POWER GRID (Guangdong, CN)
Family ID:	78999749
Assignee:	ELECTRIC POWER RESEARCH INSTITUTE, CHINA SOUTHERN POWER GRID (Guangdong, CN)
Appl. No.:	18/562310
Filed (or PCT Filed):	March 23, 2022
PCT No.:	PCT/CN2022/082445
PCT Pub. No.:	WO2023/060844
PCT Pub. Date:	April 20, 2023

Prior Publication Data

Foreign Application Priority Data

CN202111204107.4

Oct. 15, 2021

Publication Classification

Int. Cl.: H02B1/28 (20060101); H02B1/16 (20060101); H02B1/46 (20060101)

U.S. Cl.:

CPC H02B1/28 (20130101); H02B1/16 (20130101); H02B1/46 (20130101);

Field of Classification Search

CPC: H02G (3/08); H02G (3/081); H02G (3/088); H02B (1/28); H02B (1/30); H02B (1/16); H02B (1/46); H02B (1/40)

USPC: 174/50; 174/520; 174/51; 174/535; 174/559; 174/561; 174/564; 174/565; 220/3.2-; 220/4.02; 361/600; 361/601; 361/641; 361/679.01; 277/312; 277/314; 277/345

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6740809	12/2003	Vacheron	174/50	H05K 5/061
6852924	12/2004	Lessard	174/58	H05K 9/0073
8519264	12/2012	Leslie	277/652	H02G 3/088
8530760	12/2012	Taylor	174/563	H05K 5/061
8772635	12/2013	Chen	174/50	H05K 5/061
8796548	12/2013	Rost	174/50	H05K 5/061
8867235	12/2013	Veillette	174/559	H02G 3/083
8963000	12/2014	Xiao	174/548	H02G 3/16
9131609	12/2014	Ochiai	N/A	B60R 16/0239
2018/0301830	12/2017	Hoyack et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
202076590	12/2010	CN	N/A
108736188	12/2017	CN	N/A
209709469	12/2018	CN	N/A
113872061	12/2020	CN	N/A
1440922	12/1975	GB	N/A

OTHER PUBLICATIONS

International Search Report for PCT/CN2022/082445 mailed May 30, 2022, ISA/CN. cited by applicant

Background/Summary

(1) This application is the national phase of International Patent Application No. PCT/CN2022/082445, filed on Mar. 23, 2022, which claims the benefit of Chinese Patent Application No. 202111204107.4, titled “ELECTRIC POWER SAFETY AUTHENTICATION DEVICE AND SYSTEM”, filed on Oct. 15, 2021 with the China National Intellectual Property Administration, both of which are incorporated herein by reference in their entireties.

FIELD

(2) The present disclosure relates to the technical filed of auxiliary devices for power production, and in particular to an electric power safety authentication device.

BACKGROUND

(3) With city construction and economic development, the scale and quantity of power production are increasing year by year, resulting in a large number of on-site works for power production departments. For example, power enterprises perform power transmission, power distribution, and power transformation to complete construction, maintenance, and operational support of power facilities such as power transmission lines, power distribution lines, and transformer substations. In dealing with a large number of on-site production works, the power enterprises implement multiple safety mechanisms. Measures such as operation orders and work orders, designated safety personnel, and safety regulations are used to ensure safety of on-site power production works, so as to enhance safety awareness of operators and avoid casualties and device failure. Therefore, electric power safety authentication devices are essential. The safety authentication devices are usually installed in power plants. However, the conventional electric power safety authentication devices have the following advantages. 1. Water from the outside is prone to infiltrating into internal electronic components of the electric power safety authentication device due to poor waterproof performance of the electric power safety authentication device, which causes short circuits and result in incalculable losses. 2. Screws are used for connection in installation of the electric power safety authentication device, for which screw holes are not only unfavorable for waterproofing, but also prone to be oxidized and corroded, which is not conducive to installation and disassembly, and thus is not conducive to repair, maintenance, and waterproofing of the device.

SUMMARY

(4) In view of this, an objective of the present disclosure is to provide an electric power safety authentication device to effectively solve the technical problems of poor sealing and inconvenient installation in the conventional technology.

(5) In order to achieve the above objective, the following technical solutions are provided according to the present disclosure.

(6) An electric power safety authentication device includes a front casing, a rear casing, a sealing element, and a fastener connection structure. A sealing groove that matches the sealing element is provided at connection between the front casing and the rear casing. The fastener connection structure includes a first connector, a second connector, a first fastener, a second fastener, a first spring, and second springs. The first spring is arranged inside the sealing element. The first connector and the second connector are arranged at two ends of the first spring respectively, and both the first connector and the second connector penetrate the sealing element. A circular groove is provided on each of the first connector and the second connector. The first fastener is arranged on the front casing, and the second fastener is arranged on the rear casing. The first fastener and the

second fastener each include a first clamp and a second clamp. The first clamp and the second clamp are each connected to the front casing or the rear casing through the second spring. The first clamp and the second clamp collectively define a clamping space that matches the circular groove. The first connector is fit in and fastened to the first fastener, and the second connector is fit in and fastened to the second fastener.

(7) Preferably, the first connector and the second connector each include a first cylinder and a second cylinder. The circular groove is provided on the first cylinder. A first end of the second cylinder is detachably connected to the first cylinder, and a second end of the second cylinder is connected to the first spring.

(8) Preferably, the sealing element is an inflatable sealing element. The inflatable sealing element and the second cylinder are connected in a sealed manner.

(9) Preferably, the first cylinder and the second cylinder are connected through a first connecting body in threaded connection.

(10) Preferably, a threaded groove is provided on the second cylinder. The first connecting body is provided with a threaded cylinder that matches the threaded groove.

(11) Preferably, an extension-retraction groove is provided on each of a side of the first clamp and a side of the second clamp, where the two sides face away from each other. A first end of the second spring is installed in the extension-retraction groove, and a second end of the second spring is connected to the front casing or the rear casing through a connecting block.

(12) Preferably, the front casing is provided with a screen on a side away from the rear casing.

(13) Preferably, the rear casing is provided with a metal gasket on a side away from the front casing. The screen is connected to the metal gasket through a wire.

(14) Preferably, the front casing is further provided with insulated buttons on a side away from the rear casing.

(15) Preferably, the number of the fastener connection structure is more than one, and the more than one fastener connection structures are evenly distributed on the sealing element.

(16) Compared with the conventional technology, the present disclosure has the following beneficial effects. (1) In use, the first connector is fit in and fastened to the first fastener on the front casing, the second connector is fit in and fastened to the second fastener on the rear casing, and the sealing element is arranged at the connection between the front casing and the rear casing, which ensures the sealing of the connection between the front casing and the rear casing. Each of the first fastener and the second fastener is composed of the first clamp and the second clamp with an adjustable clamping distance between the first clamp and the second clamp, instead of installation through screws in the conventional technology, so that the installation in the present disclosure is convenient and simple, thereby effectively solving the technical problems of poor sealing and inconvenient installation in the conventional technology. (2) The front casing and the rear casing are installed in a sealed manner only through clamping. In addition, the sealing element has a sealing effect. The installation through screws in the conventional technology is replaced, so that installation holes are eliminated, which reduces the possibility of entry of water, thereby improving waterproof performance. (3) By setting the first spring, the distance between the first connector and the second connector can be effectively adjusted, so that the connection between the front casing and the rear casing is applicable to distribution boxes with different thicknesses, thereby effectively expanding application scope.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to describe the technical solutions in the embodiments of the present disclosure or in the conventional technology more clearly, the drawings for describing the embodiments or the

conventional technology are briefly introduced hereinafter. Apparently, the drawings in the following description show merely the embodiments of the present disclosure, and those skilled in the art can obtain other drawings based on these drawings without creative efforts.

(2) FIG. 1 is a schematic diagram showing connection between a fastener connection structure and a sealing element of an electric power safety authentication device according to an embodiment of the present disclosure;

(3) FIG. 2 is a schematic diagram showing installation of a sealing element of an electric power safety authentication device according to an embodiment of the present disclosure;

(4) FIG. 3 is a schematic diagram showing an exploded view of a fastener connection structure of an electric power safety authentication device according to an embodiment of the present disclosure;

(5) FIG. 4 is a schematic diagram showing a three-dimensional structure of an electric energy safety authentication device according to an embodiment of the present disclosure; and

(6) FIG. 5 is a schematic diagram showing installation of a metal gasket of an electrical energy safety authentication device according to an embodiment of the present disclosure.

(7) TABLE-US-00001 Reference numbers are listed as follows: 100 fastener connection structure 110 first connector 120 second connector 130 circular groove 141 first cylinder 142 second cylinder 150 first connecting body 161 threaded groove 162 threaded cylinder 210 first fastener 220 second fastener 231 first clamp 232 second clamp 240 semicircular clamping groove 310 first spring 320 second spring 330 connecting block 400 sealing element 410 sealing groove 500 front casing 510 screen 520 insulated button 600 rear casing 610 installation groove 710 first wire 720 second wire 730 metal gasket 740 conductive contact piece

DETAILED DESCRIPTION OF THE EMBODIMENTS

(8) Technical solutions in the embodiments of the present disclosure are clearly and completely described below in conjunction with the drawings of the embodiments of the present disclosure. Apparently, the embodiments described in the following are only some embodiments of the present disclosure, rather than all of the embodiments. Any other embodiments obtained by those skilled in the art based on the embodiments in the present disclosure without any creative efforts fall within the protection scope of the present disclosure.

(9) In the description of the embodiments of the present disclosure, it should be noted that the orientation or positional relationships indicated by the terms such as “central”, “upper”, “lower”, “left”, “right”, “vertical”, “horizontal”, “inner”, and “outer”, are based on the orientation or positional relationships shown in the drawings, which are only used to facilitate the description of the present disclosure and to simplify the description, rather than indicating or implying that the devices or elements referred to must have a specific orientation or can only be configured and operated in a particular orientation. Therefore, the above-mentioned terms should not be construed as limitations to the present disclosure. In addition, terms such as “first”, “second”, “third” and the like are merely for description, and should not be construed as indicating or implying relative importance.

(10) In the description of the embodiments of the present disclosure, it should be noted that, otherwise clearly specified or limited, terms such as “mounting”, “joint” and “connection” should be understood in a broad sense, for example, understood as in a fixed connection, a detachable connection or an integral connection; in a mechanical connection or an electrical connection; in a direct connection or an indirect connection through an intermediate medium, or an internal connection inside two components. For those skilled in the art, the specific meanings of the above terms in the present disclosure can be understood in the light of specific conditions.

(11) With city construction and economic development, the scale and quantity of power production are increasing year by year, resulting in a large number of on-site works for power production departments. For example, power enterprises perform power transmission, power distribution, and power transformation to complete construction, maintenance, and operational support of power

facilities such as power transmission lines, power distribution lines, and transformer substations. In dealing with a large number of on-site production works, the power enterprises implement multiple safety mechanisms. Measures such as operation orders and work orders, designated safety personnel, and safety regulations are used to ensure safety of on-site power production works, so as to enhance safety awareness of production operators and avoid casualties and device failure. Therefore, electric power safety authentication devices are essential. The safety authentication devices are usually installed in power plants. However, the conventional electric power safety authentication devices have the following advantages. (1) Water from the outside is prone to infiltrating into internal electronic components of the electric power safety authentication device due to poor waterproof performance of the electric power safety authentication device, which causes a short circuit, thereby resulting in an incalculable loss. (2) Screws are used for connection in installation of the electric power safety authentication device, for which screw holes are not only unfavorable for waterproofing, but also prone to be oxidized and corroded, which is not conducive to installation and disassembly, and thus is not conducive to repair, maintenance, and waterproofing of the device. (3) The electric power safety authentication device has poor anti-static performance and the electric power safety authentication device itself is exposed to an electric field, so that a large amount of static electricity accumulates on a surface of the electric power safety authentication device. The static electricity not only endangers a service life of the device, but also adsorbs dust and bacteria. When a human body is in contact with the safety authentication device, the bacteria contact with the human body and infect the human body, reducing resistance of the human body. An electric power safety authentication device is provided according to this embodiment to effectively solve the technical problems of poor sealing, inconvenient installation, and excessive accumulation of static electricity on the surface existing in the conventional technology.

(12) Referring to FIG. 1 to FIG. 5, an electric power safety authentication device is provided according to an embodiment of the present disclosure. The device includes a front casing **500**, a rear casing **600**, a sealing element **400**, and a fastener connection structure **100**. A sealing groove **410** that matches the sealing element **400** is provided at connection between the front casing **500** and the rear casing **600**. The fastener connection structure **100** includes a first connector **110**, a second connector **120**, a first fastener **210**, a second fastener **220**, a first spring **310**, and second springs **320**. The first spring **310** is arranged inside the scaling element **400**. The first connector **110** and the second connector **120** are arranged at two ends of the first spring **310** respectively, and both the first connector **110** and the second connector **120** penetrate the sealing element **400**. A circular groove **130** is provided on each of the first connector **110** and the second connector **120**. The first fastener **210** is arranged on the front casing **500**, and the second fastener **220** is arranged on the rear casing **600**. The first fastener **210** and the second fastener **220** each include a first clamp **231** and a second clamp **232**. The first clamp **231** and the second clamp **232** are each connected to the front casing **500** or the rear casing **600** through a second spring **320**. The first clamp **231** and the second clamp **232** collectively define a clamping space that matches the circular groove **130**. The first connector **110** is fit in and fastened to the first fastener **210**, and the second connector **120** is fit in and fastened to the second fastener **220**.

(13) Specifically, the first clamp **231** and the second clamp **232** that form the first fastener **210** are each arranged on the front casing **500** through a second spring **320**, so that the first connector **110** is connected to the front casing **500** through the first fastener **210**. The first clamp **231** and second clamp **232** that form the second fastener **220** are each arranged on the rear casing **600** through a second spring **320**, so that the second connector **120** is connected to the rear casing **600** through the second fastener **220**. The sealing element **400** and sealing groove **410** are each in a circular shape. The sealing groove **410** is provided on an edge of the front casing **500** and an edge of the rear casing **600**. The sealing groove **410** on the front casing **500** and the sealing groove **410** on the rear casing **600** collectively define a complete structure that matches the sealing element **400**, so that the

connection between the front casing **500** and the rear casing **600** is sealed with the sealing element **400**. The sealing element **400** matches the scaling grooves **410** and is arranged in the sealing grooves **410**, and the fastener connection structure **100** is arranged on the sealing element **400**, so that the sealing element **400** can position the fastener connection structure **100**, to facilitate the accurate fitting and fastening of the first connector **110** to the first fastener **210** and enables the accurate fitting and fastening of the second connector **120** to the second fastener **220**.

(14) Further, in this embodiment, the first connector **110** and the second connector **120** each include a first cylinder **141** and a second cylinder **142**. The circular groove **130** is provided on the first cylinder **141**. A first end of the second cylinder **142** is detachably connected to the first cylinder **141**, and a second end of the second cylinder **142** is connected to the first spring **310**. The first connector **110** and the second connector **120** are each divided into the first cylinder **141** and the second cylinder **142**, which facilitates processing the first cylinder **141** to form the circular groove **130** and facilitates assembling the two second cylinders **142** with the first spring **310**.

(15) More specifically, the first cylinder **141** is in a circular frustum shape at an end away from the second cylinder **142**. An inner diameter of the first cylinder **141** at an end away from the second cylinder **142** gradually decreases in a direction away from the second cylinder **142**. With such settings, it is easy for the first cylinder **141** to enter the clamping space defined by the first clamp **231** and the second clamp **232**, which facilitates the fitting and fastening of the first connector **110** to the first fastener **210** and facilitates the fitting and fastening of the second connector **120** to the second fastener **220**.

(16) Furthermore, in this embodiment, the sealing element **400** is implemented as an inflatable sealing element **400**. The inflatable sealing element **400** and the second cylinder **142** are connected in a sealed manner. The inflatable sealing element **400** may be inflated to the extent that the sealing element **400** fills all gaps at the connection between the front casing **500** and the rear casing **600**. During the inflation of the sealing element **400**, the first spring **310** is driven to extend. A continuous elastic tension force for restoring an original state of the first spring **310** causes the front casing **500** and the rear casing **600** to remain under a tight clamping force. By designing generation of two opposite forces, sealing at connection of the device is ensured, so as to ensure a sealed connection between the front casing **500** and the rear casing **600**, thereby achieving a waterproof effect.

(17) More specifically, multiple through holes, for the second cylinder **142** to protrude, are provided on the inflatable sealing element **400**. An outer wall of the second cylinder **142** and edges of the through holes are connected in a sealed manner.

(18) Furthermore, in this embodiment, the first cylinder **141** and the second cylinder **142** are connected through a first connecting body **150** via threaded connection. The threaded connection is easy to install and resistant to loosening, which not only ensures quickness in connecting the first cylinder **141** and the second cylinder **142** but also ensures stability of the connection between the first cylinder **141** and the second cylinder **142**.

(19) More specifically, the first connector **110** and the second connector **120** are symmetrically arranged with respect to a center of the second spring **320**. Such arrangement causes interchangeability between the first connector **110** and the second connector **120**, which improves convenience in use.

(20) Furthermore, in this embodiment, a threaded groove **161** is provided on the second cylinder **142**. The first connecting body **150** is provided with a threaded cylinder **162** that matches the threaded groove **161**. The connection between the second cylinder **142** and the first connecting body **150** can be hidden internally through the arrangement of the threaded groove **161** and the threaded cylinder **162**, ensuring that the second cylinder **142** and the first connecting body **150** can be connected to each other and the appearance is nice-looking.

(21) More specifically, the first connecting body **150** is connected to the first cylinder **141** through bonding or welding. A maximum outer diameter of the first cylinder **141**, a maximum outer

diameter of the second cylinder **142**, and a maximum outer diameter of the first connecting body **150** are the same. That is, surfaces of the first cylinder **141**, the second cylinder **142**, and the first connecting body **150** are flush.

(22) Furthermore, in this embodiment, an extension-retraction groove is provided on a side of the first clamp **231** and a side of the second clamp **232**, where the two sides face away from each other. A first end of the second spring **320** is installed in the extension-retraction groove, and a second end of the second spring **320** is connected to the front casing **500** or the rear casing **600** through a connecting block **330**. The first end of the second spring **320** is connected to an inner wall of the extension-retraction groove, and the second end of the second spring **320** is connected to the front casing **500** or the rear casing **600** through the connecting block **330**. Through the extension-retraction grooves, an extension-retraction range of the second spring **320** can be effectively expanded, which facilitates extension and retraction of the second springs **320** on two sides, thereby clamping and fixing the first connector **110** or the second connector **120** better.

(23) More specifically, a semicircular clamping groove **240** is provided on a side of the first clamp **231** and a semicircular clamping groove **240** is provided on a side of the second clamp **232**, where the two sides face to each other and the two semicircular clamping grooves **240** match. The two semicircular clamping grooves **240** collectively form a complete clamping groove matching an inner wall of the circular groove **130**, so that the first clamp **231** and the second clamp **232** clamp around an outer circumference of the first connector **110** or the second connector **120**, thereby achieving an effect of clamping and fixing. At least two second springs **320** are arranged on the first clamp **231**, and at least two second springs **320** are arranged on the second clamp **232**. Two second springs **320** are symmetrically arranged on the first clamp **231** or the second clamp **232**.

(24) Furthermore, in this embodiment, the front casing **500** is provided with a screen **510** on a side away from the rear casing **600**. With the screen **510**, authentication of staff through facial recognition can be achieved and non-staff whose faces not recorded cannot be authenticated, so that the staff can safely perform the electric power safety authentication operation with records.

(25) Furthermore, in this embodiment, the rear casing **600** is provided with a metal gasket **730** on a side away from the front casing **500** (i.e. the back of the rear casing **600**). The screen **510** is connected to the metal gasket **730** through a wire. An installation groove **610** is provided on the back of the rear casing **600**. The metal gasket **730** is installed in the installation groove **610**, and the metal gasket **730** is flush with the back of the rear casing **600**, which enables the metal gasket **730** to be in contact with the wall after installation. In this way, static electricity grounding is achieved through a series of circuits to successfully transfer static electricity, so as to eliminate the static electricity effect of the device through the circuits, thereby achieving a beneficial anti-static effect.

(26) More specifically, an inner wall of the front casing **500** is fixedly connected to the screen **510**. An outer wall on a side of the screen **510** is fixedly connected to a first wire **710**. An inner wall of the rear casing **600** is fixedly connected to multiple second wires **720**. Another end of the first wire **710** is electrically connected to an end of each second wire **720**, and other ends of the second wires **720** are connected to metal gaskets **730** through conductive contact pieces **740**.

(27) Furthermore, in this embodiment, the front casing **500** is further provided with insulated buttons **520** on the side away from the rear casing **600**. With the insulated buttons **520**, the staff can be authenticated through password recognition. Moreover, the insulated buttons **520** made of an insulating material can effectively avoid damage to the device and human bodies caused by accumulation of a large amount of static electricity generated by frictions in frequent use of the buttons and avoid the damage to the device caused by the accumulation of a large amount of dust due to the accumulated static electricity within the interior of the device.

(28) Furthermore, in this embodiment, the number of the fastener connection structure **100** is more than one, and the more than one fastener connection structures **100** are evenly distributed on the sealing element **400**. The connection between the front casing **500** and the rear casing **600** can be effectively strengthened through the more than one evenly distributed fastener connection

structures **100**. The sealing between the front casing **500** and the rear casing **600** can be further ensured by the sealing element **400**, which improves waterproof performance.

(29) This embodiment has the following beneficial effects. (1) In use, the first connector **110** is fit in and fastened to the first fastener **210** on the front casing **500**, the second connector **120** is fit in and fastened to the second fastener **220** on the rear casing **600**, and the sealing element **400** is arranged at the connection between the front casing **500** and the rear casing **600**, which ensures the sealing of the connection between the front casing **500** and the rear casing **600**. Each of the first fastener **210** and the second fastener **220** is composed of the first clamp **231** and the second clamp **232** with an adjustable clamping distance between the first clamp **231** and the second clamp **232**, instead of installation through screws in the conventional technology, so that the installation in the present disclosure is convenient and simple, thereby effectively solving the technical problems of poor sealing and inconvenient installation in the conventional technology. (2) The front casing **500** and the rear casing **600** are installed in a sealed manner only through clamping. In addition, the sealing element **400** has a sealing effect. The installation through screws in the conventional technology is replaced, so that installation holes are eliminated, which reduces the possibility of entry of water, thereby improving waterproof performance. (3) By setting the first spring **310**, the distance between the first connector **110** and the second connector **120** can be effectively adjusted, so that the connection between the front casing **500** and the rear casing **600** is applicable to distribution boxes with different thicknesses, thereby effectively expanding application scope. (4) By setting the screen **510** and the insulated buttons **520**, authentication can be performed by both facial recognition and composite passwords, thereby ensuring security of the authentication through dual encryption. (5) By setting the metal gasket **730** connected to the screen **510** and making the metal gasket **730** in contact with the wall after installation, static electricity grounding is achieved through corresponding circuits to successfully transfer static electricity, so as to eliminate the static electricity effect of the device through the circuits, thereby achieving a beneficial anti-static effect. (30) The above embodiments are described in a progressive manner. Each of the embodiments is mainly focused on describing its differences from other embodiments, and references may be made among these embodiments with respect to the same or similar parts. (31) With the above description of the disclosed embodiments, those skilled in the art can implement or use the present disclosure. Many modifications to these embodiments are apparent for those skilled in the art. The general principle defined herein may be applied to other embodiments without departing from the spirit or scope of the present disclosure. Therefore, the present disclosure should not be limited to the embodiments described herein but have the widest scope that complies with the principle and novelty disclosed herein.

Claims

1. An electric power safety authentication device, comprising a front casing, a rear casing, a sealing element, and a fastener connection structure, wherein the sealing element is an inflatable sealing element; a sealing groove that matches the sealing element is provided at connection between the front casing and the rear casing; the fastener connection structure comprises a first connector, a second connector, a first fastener, a second fastener, a first spring, and second springs; the first spring is arranged inside the sealing element; the first connector and the second connector are arranged at two ends of the first spring respectively, and both the first connector and the second connector penetrate the sealing element; a circular groove is provided on each of the first connector and the second connector; the first fastener is arranged on the front casing, and the second fastener is arranged on the rear casing; the first fastener and the second fastener each comprise a first clamp and a second clamp; the first clamp and the second clamp are each connected to the front casing or the rear casing through the second spring; the first clamp and the second clamp collectively define a clamping space that matches the circular groove; and the first connector is fit in and fastened to

- the first fastener, and the second connector is fit in and fastened to the second fastener.
2. The electric power safety authentication device according to claim 1, wherein the first connector and the second connector each comprise a first cylinder and a second cylinder; the circular groove is provided on the first cylinder; a first end of the second cylinder is detachably connected to the first cylinder, and a second end of the second cylinder is connected to the first spring.
 3. The electric power safety authentication device according to claim 2, wherein the inflatable sealing element and the second cylinder are connected in a sealed manner.
 4. The electric power safety authentication device according to claim 3, wherein the first cylinder and the second cylinder are connected through a first connecting body in threaded connection.
 5. The electric power safety authentication device according to claim 4, wherein a threaded groove is provided on the second cylinder; and the first connecting body is provided with a threaded cylinder that matches the threaded groove.
 6. The electric power safety authentication device according to claim 1, wherein an extension-retraction groove is provided on each of a side of the first clamp and a side of the second clamp, wherein the two sides face away from each other; and a first end of the second spring is installed in the extension-retraction groove, and a second end of the second spring is connected to the front casing or the rear casing through a connecting block.
 7. The electric power safety authentication device according to claim 1, wherein the front casing is provided with a screen on a side away from the rear casing.
 8. The electric power safety authentication device according to claim 7, wherein the rear casing is provided with a metal gasket on a side away from the front casing; and the screen is connected to the metal gasket through a wire.
 9. The electric power safety authentication device according to claim 1, wherein the front casing is further provided with insulated buttons on a side away from the rear casing.
 10. The electric power safety authentication device according to claim 1, wherein the number of the fastener connection structure is more than one, and the more than one fastener connection structures are evenly distributed on the sealing element.
-